IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Methodologies
  - Data Collection: SpaceX API and Web scraping of public websites
  - EDA: SQL and python visualizations
  - Analysis: Folium and Plotly Dash
  - Predictive Analysis: GridSearchCV, SVM,

- Results
  - Identify best launch sites and other factors that lead to successful lanches

Introduction

- SpaceY is a small startup looking to compete with SpaceX
- SpaceY needs to identify relevant factors to the success of the Falcon 9 first stage landing, so that SpaceY can correctly bit launches to customers

Section 1

Methodology

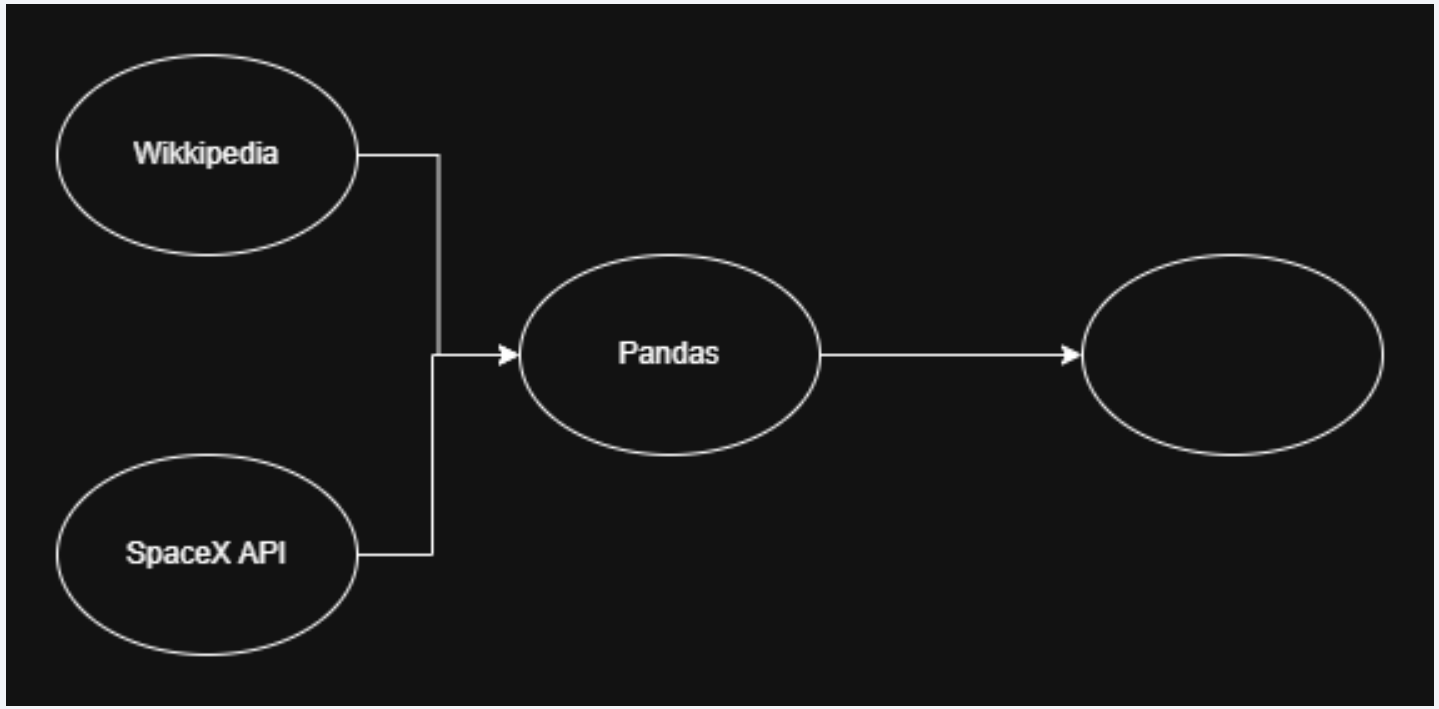
Methodology

Executive Summary

- Data collection methodology:
  - The data was collected by API requests to SpaceX and from scrapping public websites (Wikipedia)
- Perform data wrangling
  - The data was processed by using pandas SQL and Python. This mainly was used to select the Falcon 9 rockets and creating features that are more helpful to machine learning applications
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Using Sklearn's classification models and Gridsearch to find the best hyperparameters

Data Collection

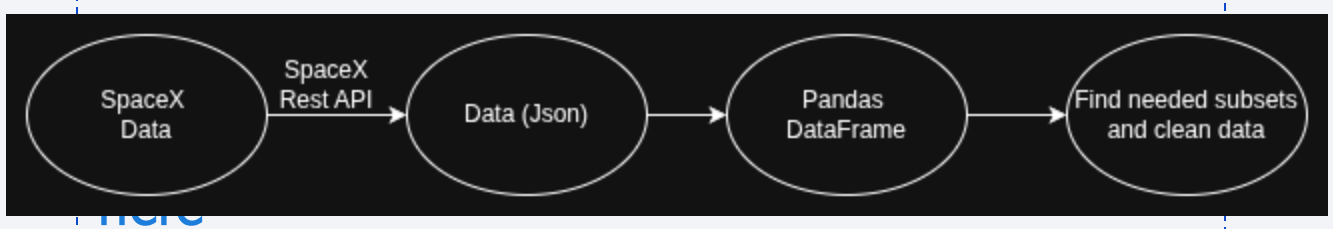
- Data was collected by web scraping Wikkipedia and using SpaceX API to load into pandas for cleaning



Data Collection – SpaceX API

- SpaceX REST calls were used to get SpaceX data that could be transformed from JSON objects into dataframes that could easily be subsetted and cleaned for a

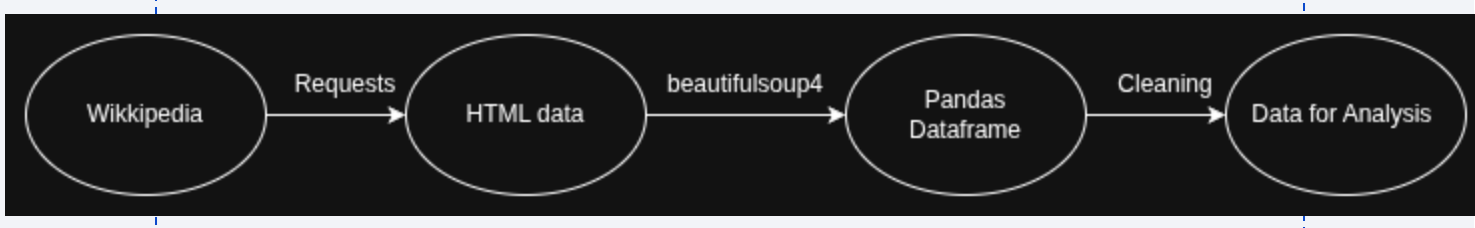
- [GitHub link]



Data Collection - Scraping

- Utilizing Python's Requests and beautifulsoup4 libraries we can programmatically get data from tables on Wikipedia and clean it for Data Analysis

- [Github URL]



Data Wrangling

- The main data transformations that occur are: filtering to the Falcon 9 booster, replacing null values with mean values, grouping success and failure into a binary variable
- Date conversion and removing unnecessary columns for analysis

- GitHub URL

EDA with Data Visualization

- Data Visualizations:
  - Flight Number vs. Launch Site
  - Payload vs. Launch Site
  - Flight Number vs. Orbit Type
  - Payload vs. Orbit Type
  - Launch Success Yearly Trend
- GitHub Link

EDA with SQL

- SQL Queries:
  - All Launch Site Names
  - Launch Site Names Begin with 'CCA'
  - Total Payload Mass
  - Average Payload Mass by F9 v1.1
  - First Successful Ground Landing Date
  - Successful Drone Ship Landing with Payload between 4000 and 6000
  - Total Number of Successful and Failure Mission Outcomes
  - Boosters Carried Maximum Payload
  - 2015 Launch Records
  - Rank Landing Outcomes Between 2010-06-04 and 2017-03-20
- GitHub URL

Build an Interactive Map with Folium

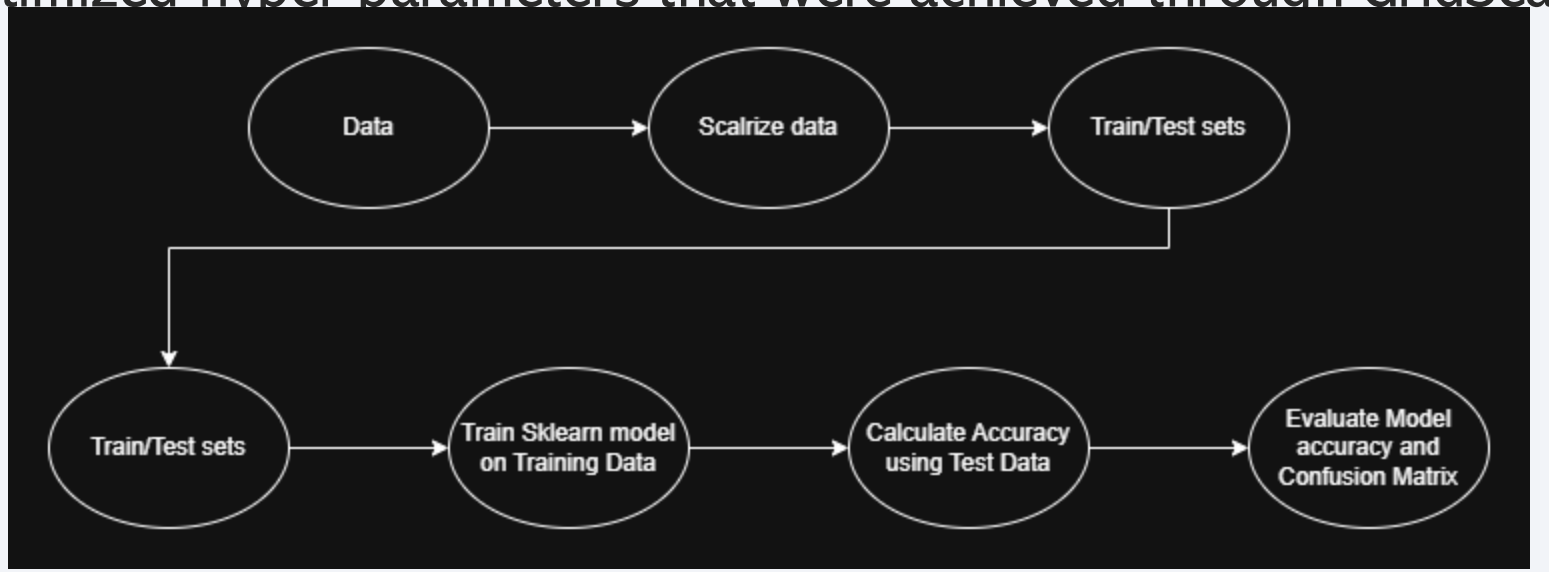
- Folium Map
  - Marked all launch sites
  - Color labeled launch outcomes
  - Proximity to coastline
- These items are crucial factors when trying to analyze and make decisions about future flight plans
- GitHub

Build a Dashboard with Plotly Dash

- Plotly Charts
  - Distribution of Successful Launches by site
  - Launch Outcome Performance for Top Site
  - Payload vs. Outcome
- These graphs highlight the insights that can be achieved through the interactive dashboard allowing for rapid iteration on ideas and finding the important correlations and points of intrest
- GitHub

Predictive Analysis (Classification)

- The predictive analysis used a randomized train and test sets while comparing different machine learning algorithms classification accuracy from optimized hyper parameters that were achieved through GridSearch



- GitHub

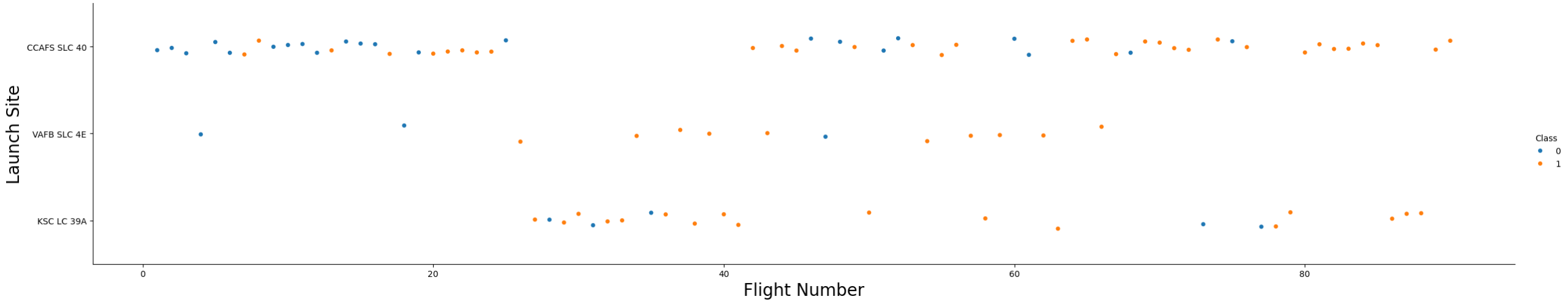
Results

- The EDA showed a strong improvement in success as more flights occurred along with an increase in payload weight. These trends suggest a high learning curve.
- The Interactive Analytics showed the importance of being near critical infrastructure such as coastline, highways, and railroads. The dashboard allowed for stakeholders to easily compare launch sites, weights, and success for quick decision making
- The Predictive Analysis showed a that modeling the success or failure of a launch could achieve an 83% accuracy rate. This allows the team to minimize the loss of expensive hardware

Section 2

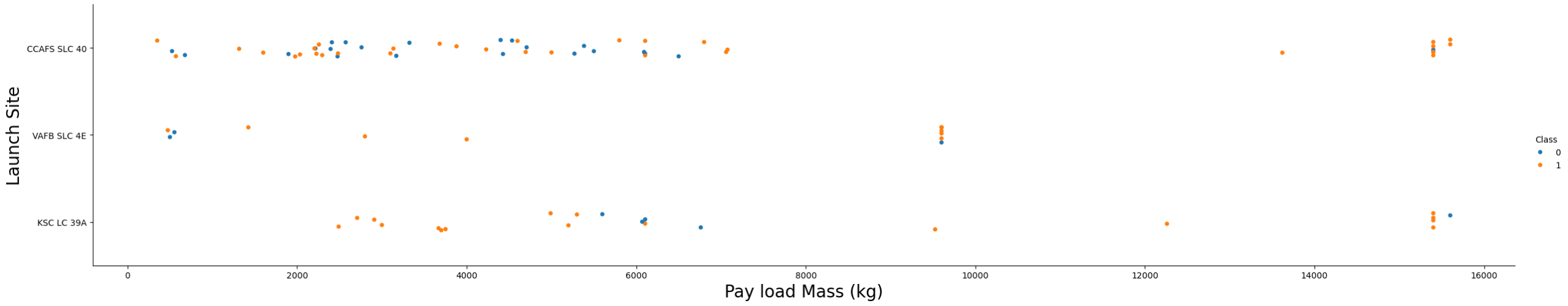
Insights drawn from EDA

Flight Number vs. Launch Site



- This graph illustrates the relationship between launch site, flight number and success, with a general trend of success the more flights are tried from each site. This shows a steep learning curve for each site

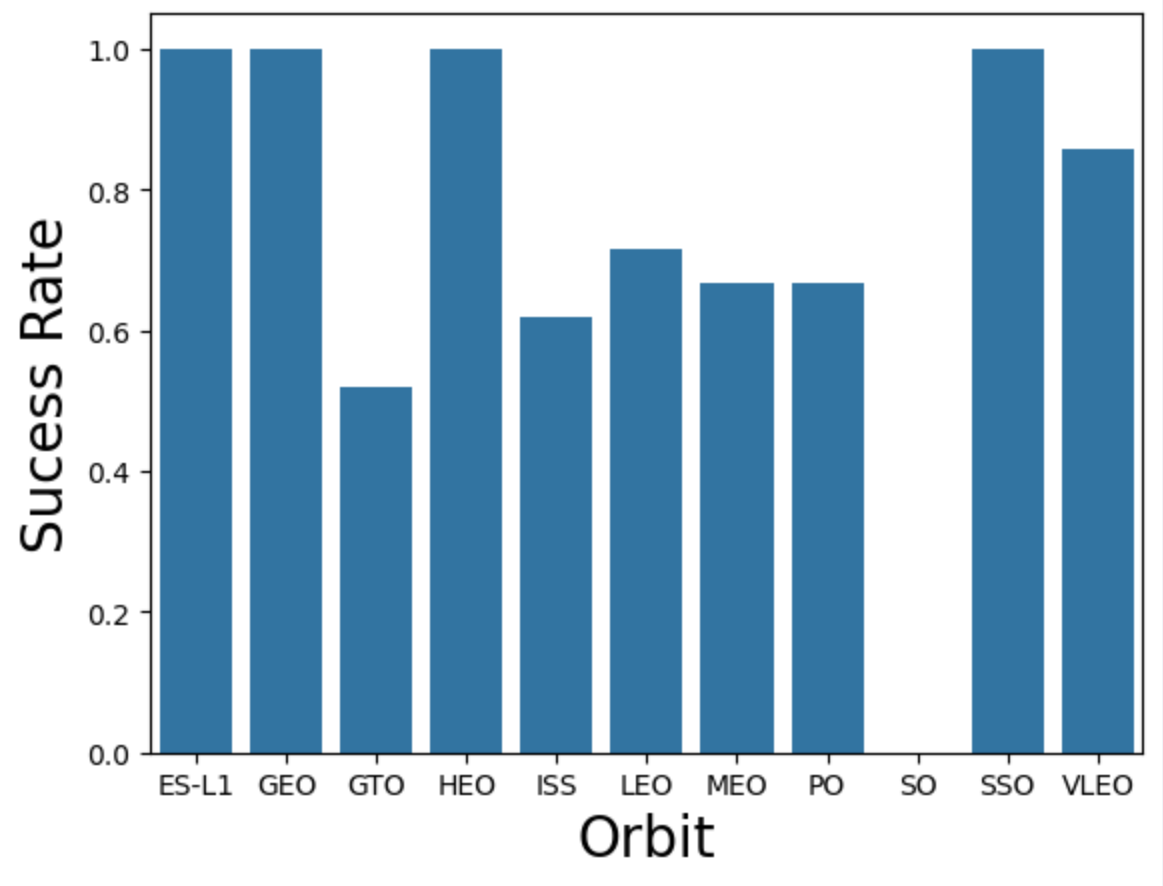
Payload vs. Launch Site



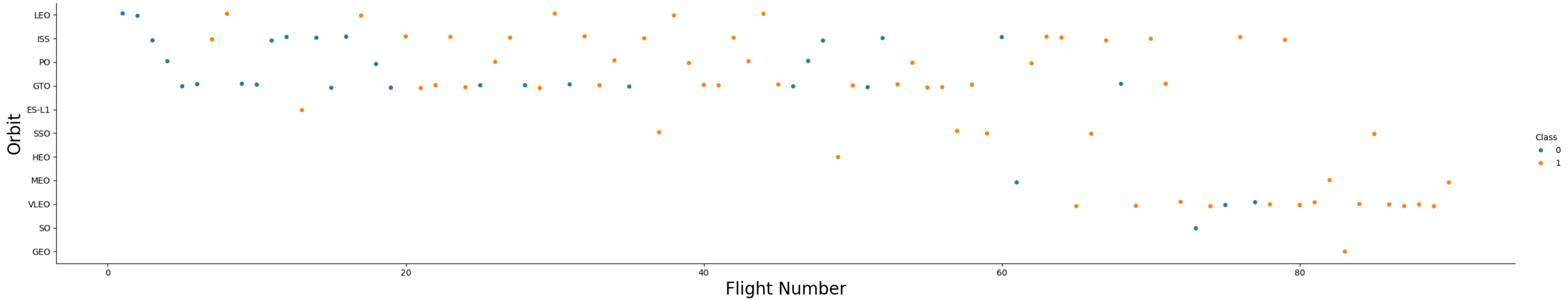
- This graph illustrates the relationship between the launch site, payload mass, and success
- The trend of higher masses being more successful at each location is an important parameter to remember for future work

Success Rate vs. Orbit Type

- The success rate of each orbit allows SpaceY to make strategic decisions about bidding prices for different orbits

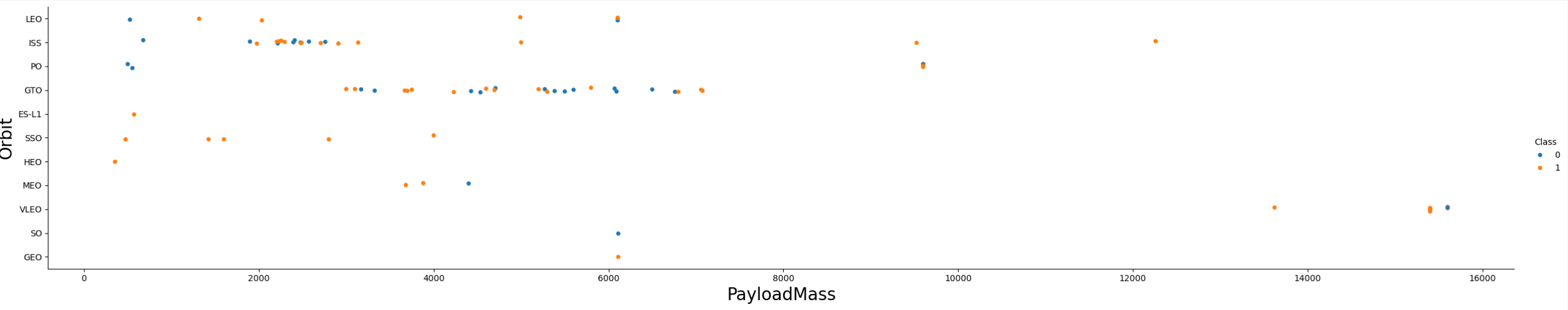


Flight Number vs. Orbit Type



- This helps us visulized how orbits and success have changed over time (flight number)
- This highlights the general trend of more successes as more flights happen but also shows they have moved to be primarily VLEO orbital launches

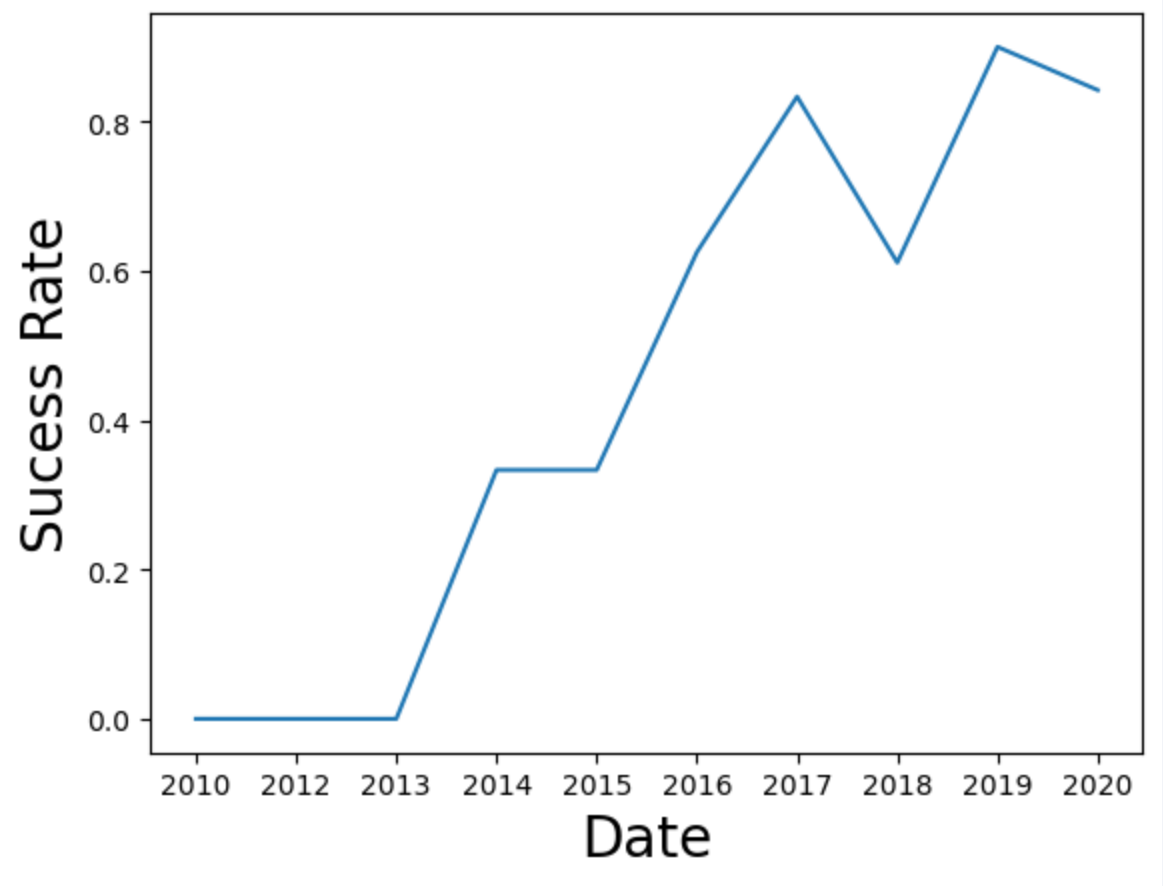
Payload vs. Orbit Type



- This helps illustrate the relationship between orbit and payload mass showing how some orbits tend to have heavier or lighter payloads and how both can affect success

Launch Success Yearly Trend

- The yearly trend of success rate highlights the steep learning curve or landing the booster rockets. This is an incredibly challenging problem and there should be precautions taken.



All Launch Site Names

- Quick query to find all distinct launch sites

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

5 records that launch site name begins with 'CCA'

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- The total payload mass carried for NASA

sum("PAYLOAD_MASS__KG_")
45596

Average Payload Mass by F9 v1.1

- The average payload mass carried by a F9 v1.1 is 2928.4, this is import to know when designing a competing booster.

avg("PAYLOAD_MASS__KG_")
2928.4

First Successful Ground Landing Date

- The first date a successful Ground landing occurred

min("Date")
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- The Booter versions that have landed a payload between 4000 and 6000 Kilograms

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

The count of Mission Outcomes

Mission_Outcome	count(*)
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- Booter versions that have carried the maximum payload

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- Failed launches from 2015 by month.
- There were relatively few failures in 2015

month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Landing outcome between 2010-06-04 and 2017-03-20

Landing_Outcome	count(*)
Failure (parachute)	31

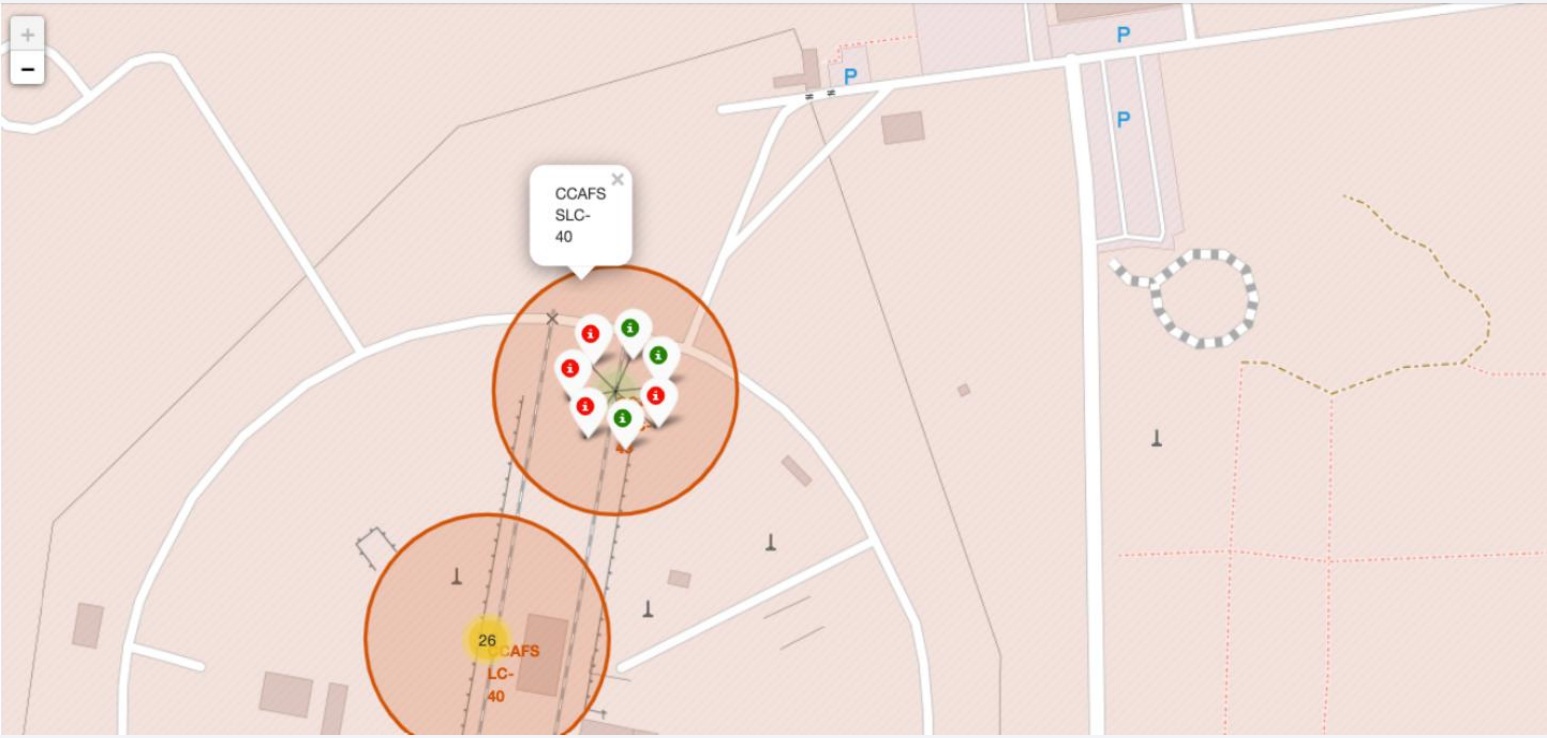
Section 3

Launch Sites Proximities Analysis

Launch Sites

- From this map you can see the importance of being near the equator and coastline

Launch Sites Labeled by Color



- Folium here lets us look at launch sites and visually see the successful and unsuccessful launches

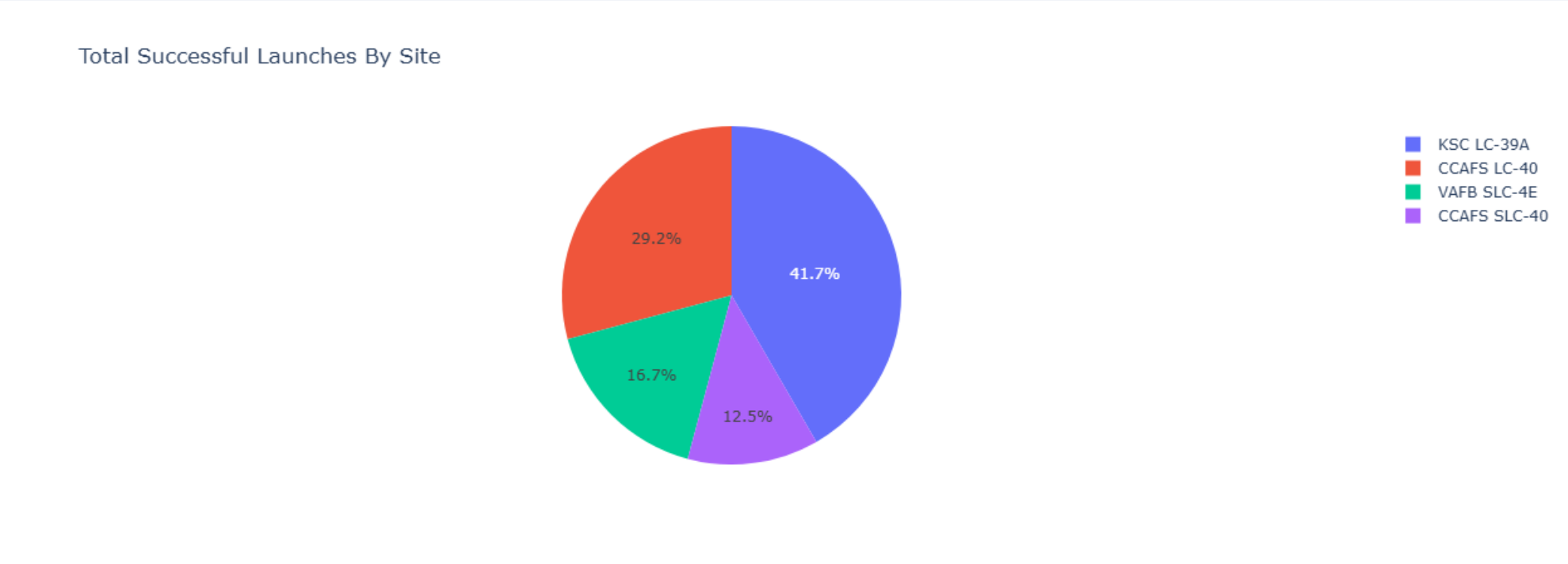
Proximity to Coastline

- The proximity to the coastline helps with water recovery and placement by ship

Section 4

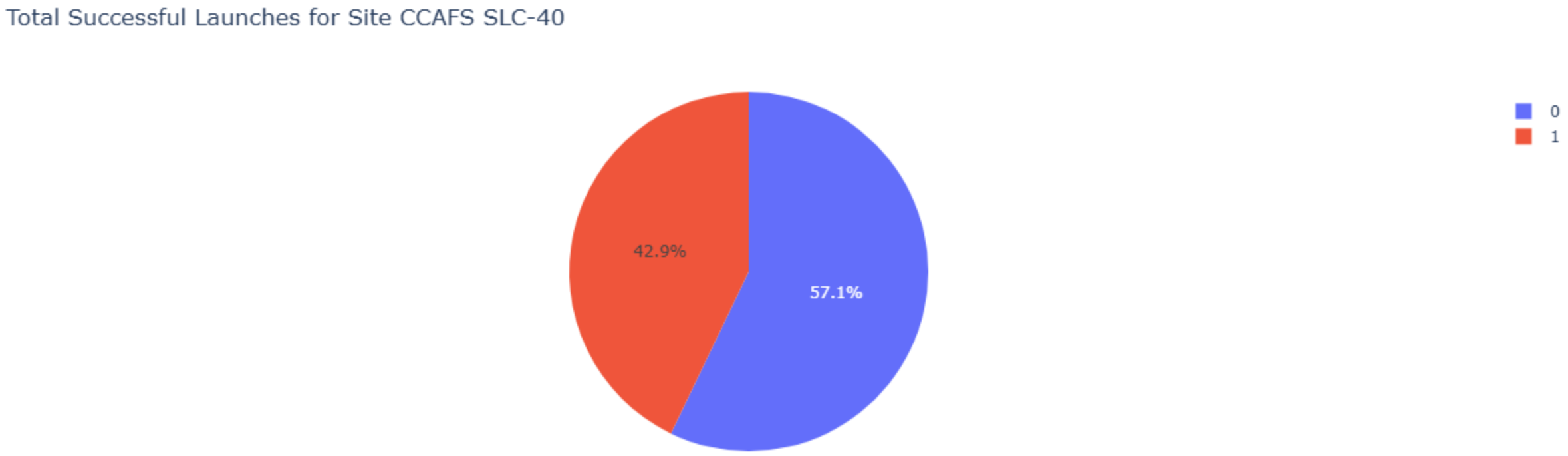
Build a Dashboard with Plotly Dash

Successful Landings of all Sites



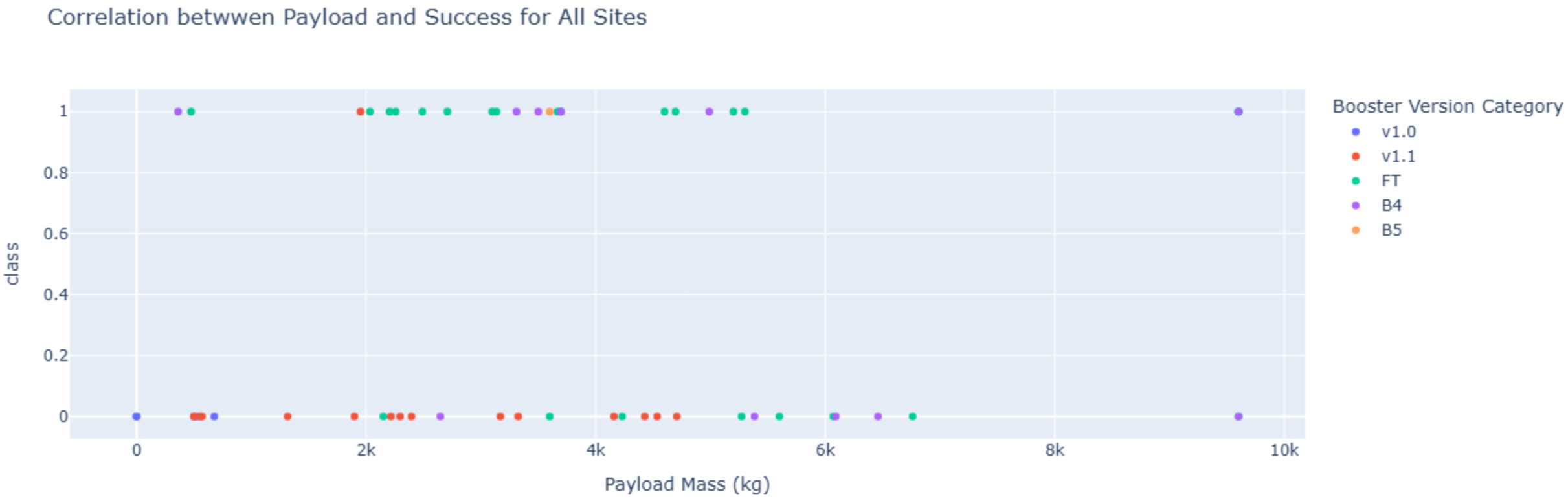
- This graph shows us which site is responsible for the most successful landings

Most Successful Site



- This shows us how often the most successful site has a successful Landing

Success vs Payload by Booster Version



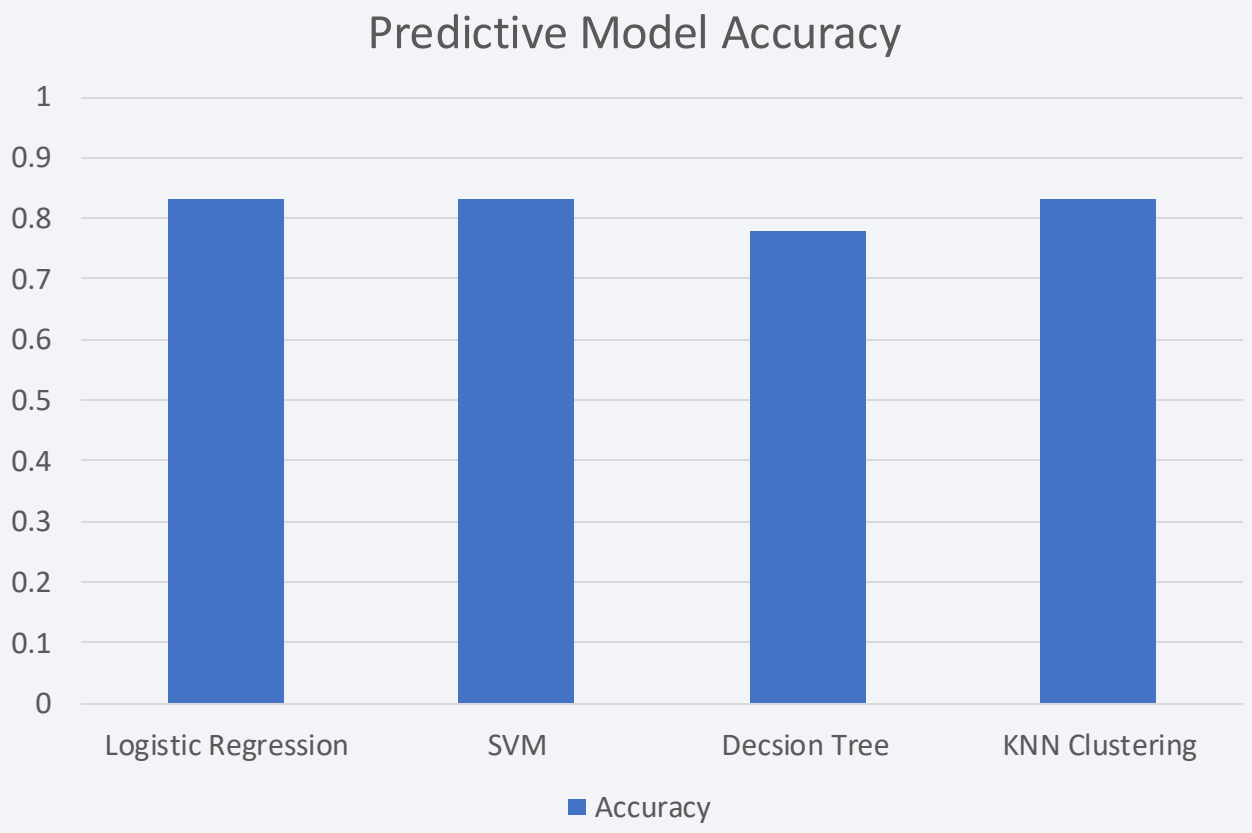
- This graph illustrates the payload and booster version for successful flights at all launch sites

Section 5

Predictive Analysis (Classification)

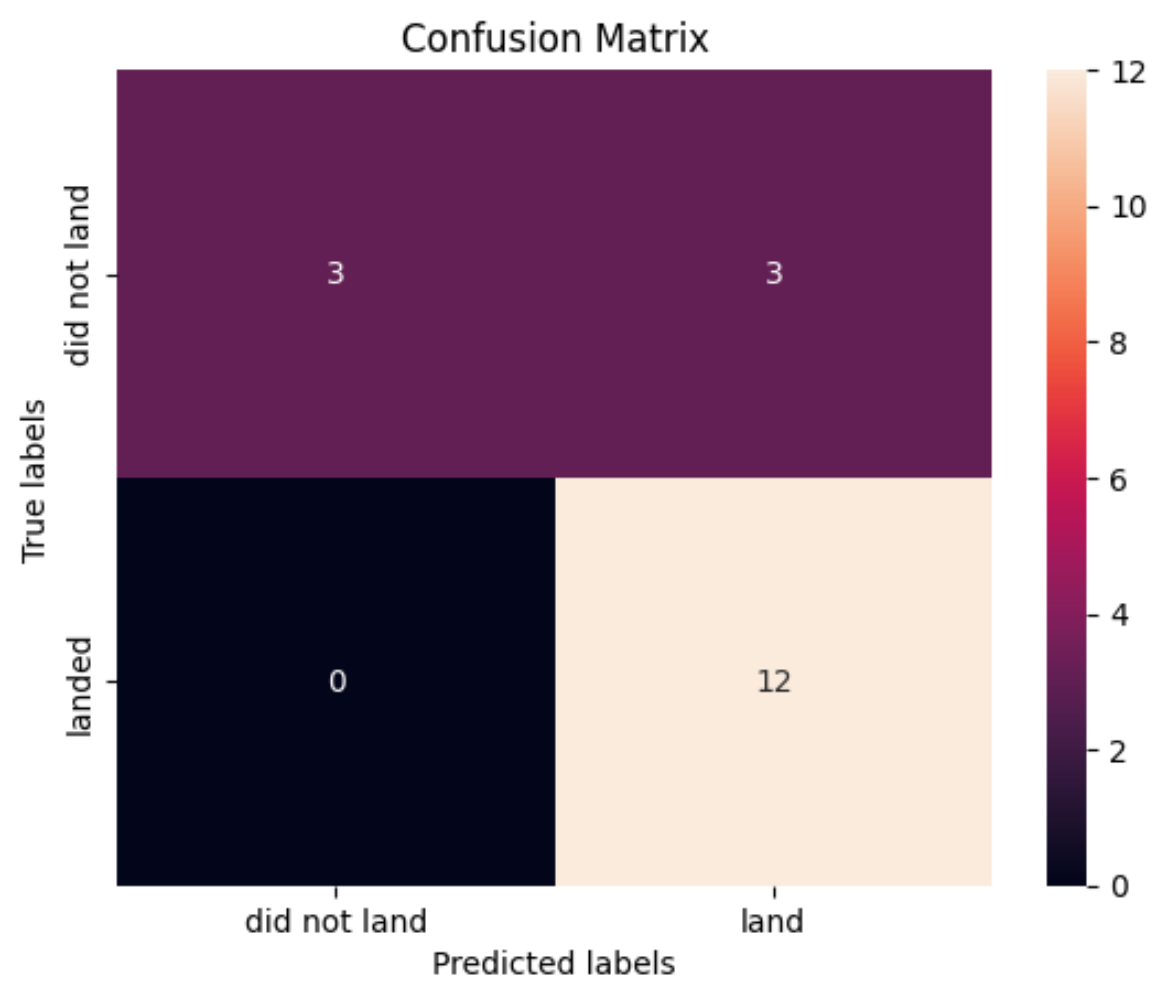
Classification Accuracy

- I had 3 models tied for most accurate and I would choose logistic regression for its simplicity, understandability and ease of use



Confusion Matrix

- The confusion matrix shows an accurate and well preforming model that does have few false negatives but overall, a very good result



Conclusions

- EDA found pattern and relationships to explore while giving a focus to the next steps
- Visual Analytics helped explore the patterns shown before and then make decisions based off the data
- Predictive Analysis gave us a place helped us understand what level of success we can hope to achieve and where to focus our efforts to get there

Thank you!